INTRODUCTION

We come across many organic compounds containing a group in our everyday Life. Biomolecules such as protein, carbohydrate etc… that makeup all plants and animals contains carbonyl group. They play an important role in the metabolic process. For example, pyridoxal, an aldehyde derived from vitamin B, function as a co –enzyme. Carbonyl compounds are important constituents of fabrics, plastis and drugs. For example, Formaldehyde is used for the manufacture of Bakelite and paracetamol, (p– acetylated aminophenol) a drug used to reduce fever, contains a carbonyl group. In this unit, we will learn the preparation, properties and uses of aldhydes, ketones and carboxylic acids.

12.1 Nomenclature of Aldehydes and ketones

We have already learnt the IUPAC system of nomenclature of organic compounds in XIth standard. Let us apply the rules to name the following compounds.

Compound (common name, Structural formula, IUPAC Name)	IUPAC Name			
	Prefix with position number	Root used	Primary suffix	Secondary Suffix
Formaldehyde H – CHO methanal	–	meth	an¢	al
Acetaldehyde CH_3 – CHO ethanal	–	eth	an¢	al
Acrolein $CH_2 = CH – CHO$ prop – 2- enal	–	prop	2-en¢	al
Crotonaldehyde $CH_3 – CH = CH – CHO$ but – 2 – enal	–	but	2-en¢	al
Glyceraldehyde $HO — CH_2 — \underset{\displaystyle OH}{\underset{|}{CH}} — CHO$ 2, 3 – dihydroxy propanal	2, 3 dihydroxy	prop	an¢	al

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